IGCSE 0607 Practice Questions — E3.7

The logarithmic function

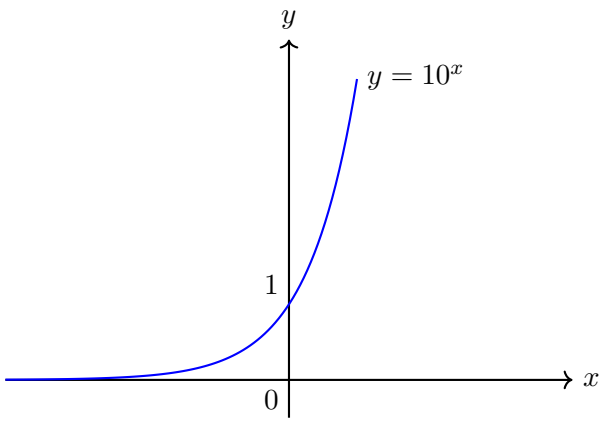
Total Questions: 39 — **Total Marks:** 235 Marks

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- **E3.7** The logarithmic function
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(26 Qs)*

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1 The diagram shows a sketch of the function $f(x) = 10^x$.



(a) Write down the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

(b) Work out the exact value of $f^{-1}\left(\frac{1}{100}\right)$.

..... **[1]**

(c) Write down the exact solution to the equation $f(x) = 45$.

$x =$ ..... **[1]**

2 **(a)** Write $10^3 = 1000$ as a logarithmic equation.

..... **[1]**

(b) Find the exact value of x when $\log x = -2$.

$x =$ ..... **[1]**

(c) Solve $5^x = 12$, giving your answer in the form $\dfrac{\log a}{\log b}$.

$x =$ ..... **[1]**

3 The functions f and g are defined as $f(x) = 10^x$ and $g(x) = \log x$.

(a) Find the value of $g(100)$.

..... **[1]**

(b) Find the value of x when $f(x) = 0.001$.

$x =$ ..... **[1]**

(c) Write down the expression for $g(f(x))$ in its simplest form.

$g(f(x)) =$ ..... **[1]**

4 **(a)** Work out the value of

$\log 10000$

..... **[1]**

(b) Write the following equation in the form $\log_a b = c$.

$5^3 = 125$

..... **[1]**

(c) Solve the equation

$\log_2 x = 6$

$x =$ ..... **[1]**

5 **(a)** Find the value of x when $\log_2 x = 5$.

$x =$ ..... **[1]**

(b) Find the value of y when $10^y = 0.01$.

$y =$ ..... **[1]**

(c) Write the equation $P = k^t$ in logarithmic form.

..... **[1]**

6 **(a)** Write the following as a logarithmic equation.

$5^3 = 125$

..... **[1]**

(b) Find the value of $\log 1000$.

..... **[1]**

(c) Solve the equation.

$\log_2 x = 5$

$x =$ ..... **[2]**

7 **(a)** Write the equation $10^5 = 100000$ in the form $\log a = b$.

..... **[1]**

(b) Find the value of x when $\log x = 3$.

$x =$ ..... **[2]**

(c) The function $f(x)$ is defined as $f(x) = 10^x$.
Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

8 All logarithms in this question are in the base shown, or base 10
10 where no base is written.

(a) Write down the value of

(i) log
1000
log1000,

..... **[1]**

(ii) log
2
32
log
2
32.

..... **[1]**

(b) Given that
log
3
=
4
log
3
x=4, find the value of
x.

=
x= ..... **[1]**

(c) The function
f is defined by
(
)
=
10
f(x)=10
x
.
Write down
1
(

)
f
1
(x).

1
(
)
=
f
1
(x)= ..... **[1]**

9 **(a)** Find the value of x in each of the following equations.

(i) $10^x = 100000$

$x =$ ..... **[1]**

(ii) $\log x = -2$

$x =$ ..... **[1]**

(iii) $\log_2 64 = x$

$x =$ ..... **[1]**

(b) Given that $f(x) = 10^x$, write down an expression for $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

10 **(a)** Write down the value of

$\log 1000$

..... **[1]**

(b) Solve the equation.

$\log x = -2$

$x =$ ..... **[2]**

(c) Write the equation $5^x = 125$ in the form $x = \log_a b$.

$x =$ ..... **[1]**

11 Answer the following independent questions on logarithmic and exponential equations.

(a) Write the exponential equation $10^4 = 10000$ in logarithmic form.

..... **[1]**

(b) Find the exact value of x when $\log_5 x = 3$.

$x =$ ..... **[2]**

(c) The solution to the equation $7^x = 25$ can be written in the form $x = \dfrac{\log b}{\log a}$.
Write down the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

12 Consider the function $f(x) = \log_3 x$ for $x > 0$.

(a) Find the value of $f(81)$.

..... **[1]**

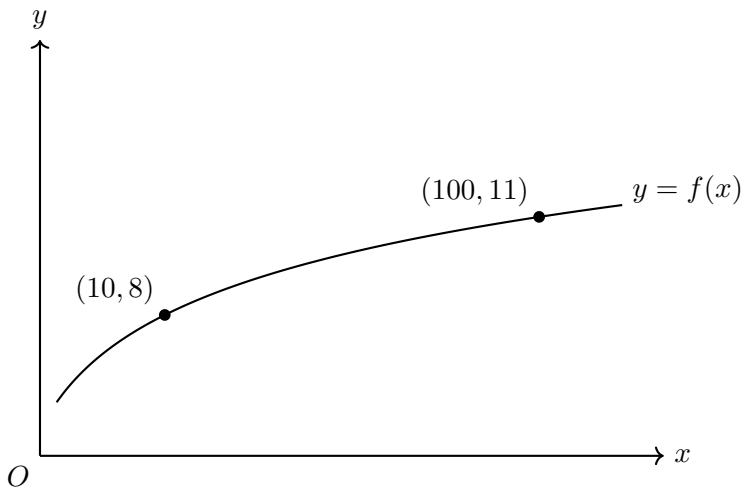
(b) Find the value of $f\left(\frac{1}{9}\right)$.

..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

13 The diagram shows a sketch of the graph of $y = f(x)$ for $x > 0$.
The function is defined as $f(x) = p \log x + q$, where $\log x$ denotes the logarithm to base 10.
The graph passes through the points $(10, 8)$ and $(100, 11)$.



(a) **(a)** Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(b) Solve the equation $f(x) = 23$.

$x =$ ..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

14 The logarithmic function is the inverse of the exponential function.

(a) Write the equation $y = 6^x$ in logarithmic form.

(a) Write the equation $y = 6^x$ in logarithmic form.

$x =$ ..... **[1]**

(b) Solve the equation.

$3^x = 20$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

15 A population of insects is growing exponentially.
The number of insects, N, after t days is given by

$$N = 250 \times 1.08^t$$

(a) Work out the number of insects after 10 days.
Give your answer to the nearest integer.

..... **[2]**

(b) Calculate the number of days it takes for the population to reach 1000.
Give your answer correct to 1 decimal place.

..... days **[3]**

16 The value of a machine, V dollars, depreciates exponentially according to the formula

$$V = 15000 \times 0.8^t$$

where t is the number of years after it is purchased.

(a) Calculate the value of the machine after 4 years.

$ ..... **[2]**

(b) Find the number of years it takes for the value of the machine to decrease to $7500.
Give your answer correct to 3 significant figures.

..... years **[3]**

17 A machine is purchased for \$25000. Its value depreciates exponentially by 12% each year.

(a) Calculate the value of the machine after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[2]**

(b) Find the number of complete years it takes for the value of the machine to fall below \$10000.
You must show all your working.

..... years **[3]**

18 **(a)** Solve the equation.

$3^{2x} = 50$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$y = 10^{x-4}$

$x =$ ..... **[2]**

19 A population of bacteria is modelled by the formula

$P = 5000 \times 1.12^t$ where P is the population and t is the time in hours after the start of an experiment.

(a) **(a)** Write down the initial population of the bacteria.

..... **[1]**

(b) Find the time it takes for the population to reach 20 000.
Give your answer correct to 1 decimal place.

$t =$ ..... hours **[4]**

20 An investment of \$8000 earns compound interest at a rate of 3.5% per year. The value of the investment after n years is \$$V$.

(a) Write down an equation for V in terms of n.

$V =$ ..... **[1]**

(b) Calculate the minimum number of complete years it takes for the value of the investment to exceed \$12000.
Show all your working.

..... years **[3]**

(c) Solve the equation.

$2^x = 55$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

21 **(a)** Solve the equation $3^x = 20$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) The mass, M grams, of a radioactive substance after t days is given by the formula
$M = 500 \times 0.85^t$ Calculate the value of t when $M = 100$.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

22 The value, V dollars, of a painting t years after it was bought is given by the formula

$$V = 4500 \times 1.08^t$$

(a) Calculate the value of the painting 5 years after it was bought.
Give your answer correct to the nearest dollar.

..... **[2]**

(b) Calculate the time it takes for the value of the painting to reach \$10 000.
Give your answer in years, correct to 2 decimal places.

..... years **[4]**

23 The value, $\$V$, of a vintage motorcycle t years after it was purchased is modelled by the formula

$$V = 4500 \times 1.08^t$$

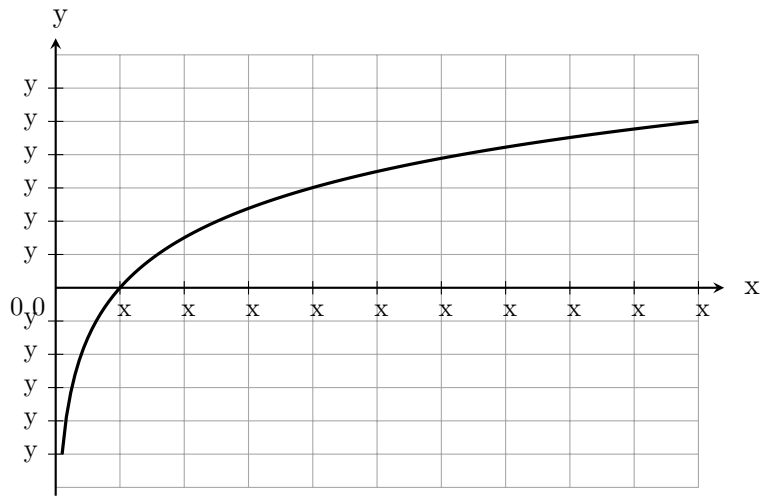
(a) Calculate the value of the motorcycle after 5 years.

$\$$ ..... **[2]**

(b) Calculate the number of years it takes for the value of the motorcycle to reach \$10 000. Give your answer correct to 1 decimal place.

..... years **[4]**

24 The diagram shows the graph of
=
log
y=logx for
0.1
10
0.1x10.



(a) Use the graph to estimate the value of
log
6
log6.

..... **[1]**

(b) Use the graph to solve the equation
log
=
0.5
logx=0.5.

=
x= ..... **[1]**

(c) Solve the equation
3
+
1
=
20
3
x+1
=20, giving your answer correct to

3
3 significant figures.

=
x= . . . . . **[3]**

(d) Solve the equation
log
(
2
3
)
=
1.4
log(2x3)=1.4, giving your answer correct to
3
3 significant figures.

=
x= . . . . . **[2]**

25 The mass, M grams, of a radioactive substance after t days is given by the formula $M = 400 \times 0.88^t$.

(a) Calculate the mass of the substance after 5 days.
Give your answer correct to 1 decimal place.

..... grams **[2]**

(b) Find the number of days it takes for the mass of the substance to reduce to exactly half of its initial value.

..... days **[3]**

(c) Another radioactive substance has an initial mass of 800 grams.
After 10 days, its mass is 300 grams.
It decays according to the formula $M = 800 \times k^t$.
Find the value of k, giving your answer correct to 3 significant figures.

$k =$ ..... **[3]**

26 An amount of \$8000 is invested in an account that pays compound interest at a rate of 4.5% per year. The value of the investment, V dollars, after t years is given by the formula

$$V = 8000 \times 1.045^t$$

(a) Calculate the value of the investment after 5 years.

\$ ..... **[2]**

(b) Use logarithms to calculate the number of complete years it takes for the value of the investment to first exceed \$15000.

..... years **[3]**

(c) Solve the equation.

$7^{x-1} = 250$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

27 A population of insects grows exponentially.
The number of insects, N, after t weeks is given by the formula

$$N = 800 \times 2^t$$

(a) Work out the number of insects after 4 weeks.

..... **[1]**

(b) The number of insects reaches 102 400.
Work out the exact value of t.

$t =$ ..... **[2]**

(c) Another population of insects can be modelled by the formula $P = 1500 \times 3^t$.
Find the value of t when this population reaches 20 000.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

(d) A sum of \$2000 is invested at an annual compound interest rate of 5%. The total value, V, of the investment after n years is given by

$$V = 2000 \times 1.05^n$$

Calculate the minimum number of complete years required for the investment to double in value.

..... years **[3]**

28 A function f is defined as $f(x) = 4 \times 10^{2x} - 7$.

(a) Find $f^{-1}(x)$.

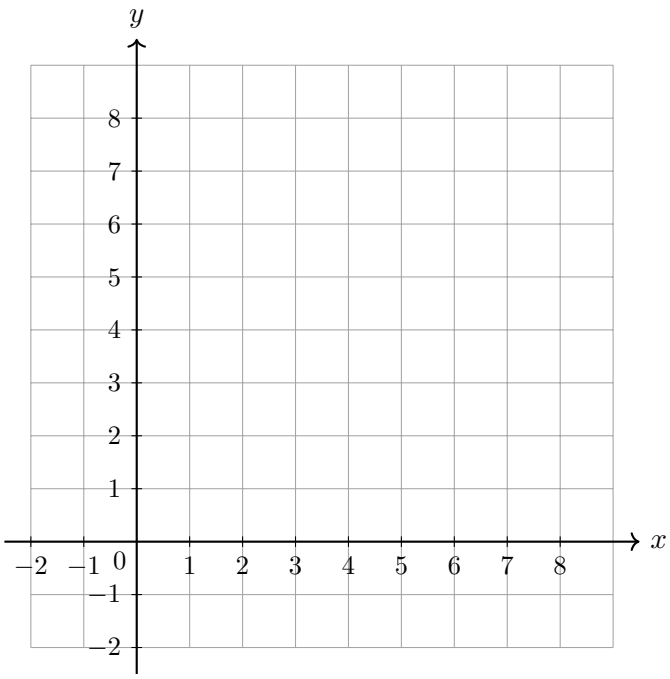
$f^{-1}(x) =$ ..... **[3]**

(b) Solve the equation $f(x) = 53$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

29 A function is defined as $f(x) = 2^x$.

(a) On the grid, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 3$.



[2]

(b) Write down the equation of the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

(c) On the same grid in part (a), sketch the graph of $y = f^{-1}(x)$ for $0 < x \leq 8$. **[2]**

(d) Solve the equation $2^x = 12$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

30 **(a)** The mass, M grams, of a radioactive substance after t days is given by $M = 250 \times 0.92^t$.

(i) Write down the initial mass of the substance.

..... grams **[1]**

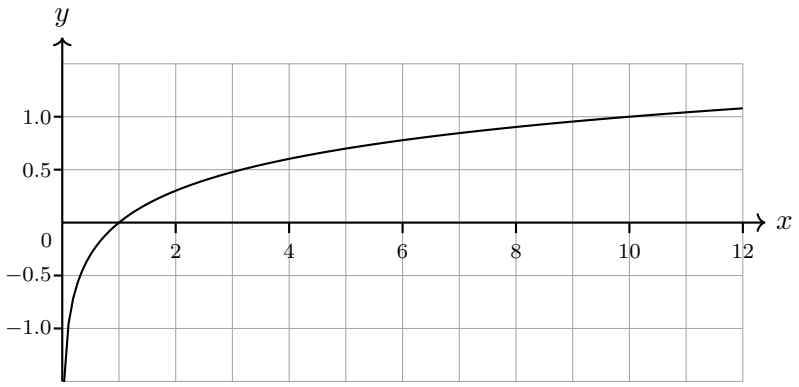
(ii) Find the number of days it takes for the mass to reduce to 50 grams.
Give your answer correct to 3 significant figures.

..... days **[3]**

(b) An amount of \$800 is invested at a rate of 6% per year compound interest. Calculate the number of complete years it takes for the investment to first exceed \$2000.

..... years **[3]**

31 The diagram shows the graph of $y = \log x$ for $0 < x \leq 12$, where $\log x$ denotes $\log_{10} x$.



(a) Write down the coordinates of the x-intercept and the equation of the vertical asymptote of the graph of $y = \log x$.

x-intercept = ... , Asymptote: ..... **[2]**

(b) Use the graph to estimate the solution to the equation $\log x = 0.7$.

x = ... **[1]**

(c) Solve algebraically the equation

$$3^{2x-1} = 5^{x+2}$$

Give your answer correct to 3 significant figures.

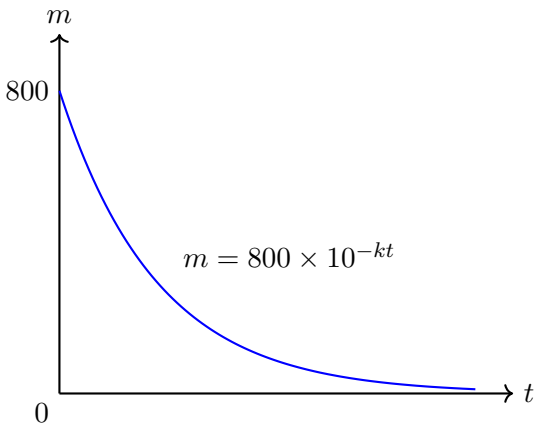
x = ... **[4]**

32 The mass, m grams, of a radioactive isotope decays according to the formula

$m = 800 \times 10^{-kt}$ where t is the time in years and k is a constant.

The diagram shows a sketch of the decay curve.

After 15 years, the mass is 250 grams.



(a) Show that $k = 0.0337$ correct to 3 significant figures.

[3]

(b) Find the number of years it takes for the mass to become 50 grams.

..... years **[3]**

(c) Make t the subject of the formula $m = 800 \times 10^{-kt}$.

$t =$ ..... **[2]**

33 Let $f(x) = 5 \times 10^x$ and $g(x) = \log x$.

(a) Find the value of $f(g(100))$.

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Solve the equation.

$f(x) = 200$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

(d) Find $g(f(x))$ in the form $x + k$, where k is a constant.

..... **[2]**

34 A function is defined as $f(x) = 5 \times 3^x - 2$.

(a) Find the value of $f(-2)$, giving your answer as a fraction in its simplest form.

..... **[2]**

(b) Find $f^{-1}(x)$.

Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

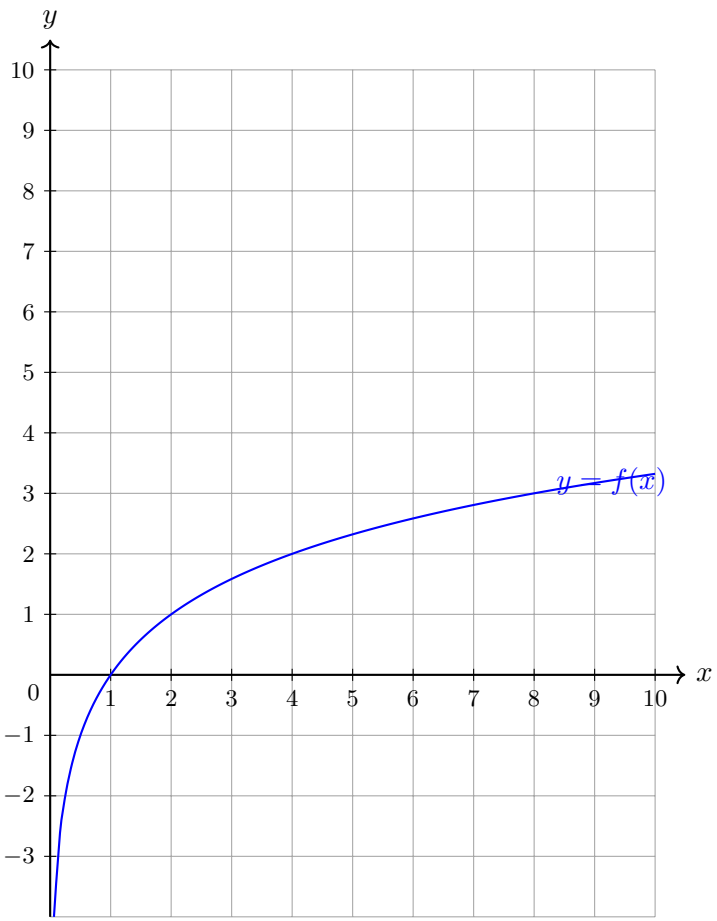
(c) Solve the equation.

$f(x) = 500$

Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

35 The diagram shows the graph of $y = f(x)$.



(a) The graph of $y = f(x)$ passes through the point $(8, 3)$.
Given that $f(x) = \log_a x$, show that $a = 2$.

[2]

(b) **(i)** On the grid, sketch the graph of $y = f^{-1}(x)$. **[2]**

(ii) Write down the equation of the asymptote of the graph of $y = f^{-1}(x)$.

..... **[1]**

(c) By drawing a suitable straight line on the grid, find an approximate solution to the equation $\log_2 x + x = 8$.

$x =$ ..... **[3]**

36 The function $f(x)$ is defined as $f(x) = 4 \times 10^{0.5x}$.

(a) Find the value of $f(3)$.
Give your answer correct to 3 significant figures.

..... **[2]**

(b) Find $f^{-1}(x)$.

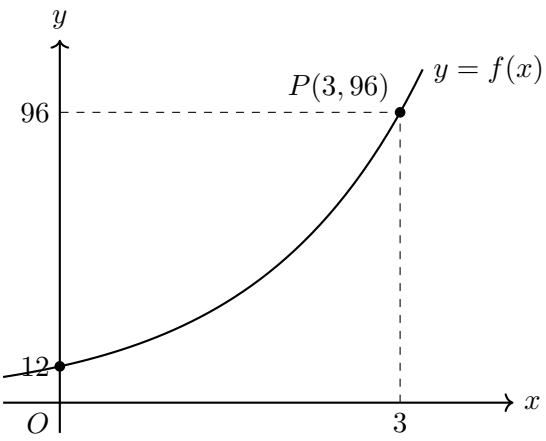
$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation

$$f(x) = 200$$

$x =$ ..... **[3]**

37 The diagram shows the graph of $y = f(x)$ where $f(x) = a \times b^x$.
The graph intersects the y-axis at the point $(0, 12)$ and passes through the point $P(3, 96)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(b) Find $f^{-1}(x)$.
Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation $f(x) = 500$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

38 The value, \$V, *of a machine t years after it is purchased is modelled by the formula* $V = 24000 \times 0.85^t$.

(a) Calculate the value of the machine after 4 years, giving your answer correct to the nearest dollar.

\$ ..... **[2]**

(b) Find the number of years it takes for the value of the machine to fall to \$9000.
Give your answer correct to 3 significant figures.

$t =$ ..... years **[3]**

(c) A different asset increases in value by 6.8% each year.
Find the minimum whole number of years it takes for the value of this asset to double.

..... years **[3]**

(d) The function g is defined by $g(x) = 5 \times 2^x + 3$.
Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

39 The function $f(x) = A \times b^x$, where $b > 0$, passes through the points $(0, 7)$ and $(3, 56)$.

(a) Find the value of A and the value of b.

$A =$ ..... , $b =$ ..... **[3]**

(b) Find the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(c) The equation $f(x) = 600$ is equivalent to $2^x = k$.

(i) Work out the exact value of k as a fraction.

$k =$ ..... **[1]**

(ii) Solve the equation $f(x) = 600$.
Give your answer correct to 3 decimal places.

$x =$ ..... **[2]**

(d) A different function $g(x)$ is defined as $g(x) = \log_c(x)$.
The graph of $y = g(x)$ passes through the point $(81, 4)$.

(i) Find the value of c.

$c =$ ..... **[2]**

(ii) Solve the equation $g(x) = -2$.

$x =$ ..... **[1]**